IAM HOME LAB PORTFOLIO

# Lab 01

## Active Directory Environment Setup & On-Premises Identity Operations

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|---|---|
| **Lab Number** | Lab 01 of 20 |
| **Phase** | Phase 1 — Active Directory & On-Premises Identity Operations |
| **Duration** | Weeks 1–2 |
| **Environment** | VMware Workstation • Windows Server 2022 • Active Directory |
| **Status** | Completed ✓ |
| **Date Completed** | April 1, 2026 |
| **Domain** | FictionalCorp.local |
| **Server Name** | DC01 (Domain Controller) |

## 1. Lab Objective

This lab establishes the on-premises Active Directory foundation for the FictionalCorp hybrid identity environment. The objective is to simulate the real-world setup that an IAM Analyst inherits or builds when joining an organization a properly structured domain with role-based Organizational Units, security groups, and provisioned user accounts managed through a documented ticketing workflow.

By completing this lab, the following competencies are demonstrated:

- Deploying and configuring a Windows Server 2022 Domain Controller
- Designing an OU structure that reflects real enterprise department organization
- Creating role-based security groups following the principle of least privilege
- Processing provisioning requests from mock HR tickets
- Applying contractor account lifecycle controls including automatic expiry
- Maintaining an audit-ready provisioning log for every account action

## 2. Lab Environment

### 2.1 Infrastructure

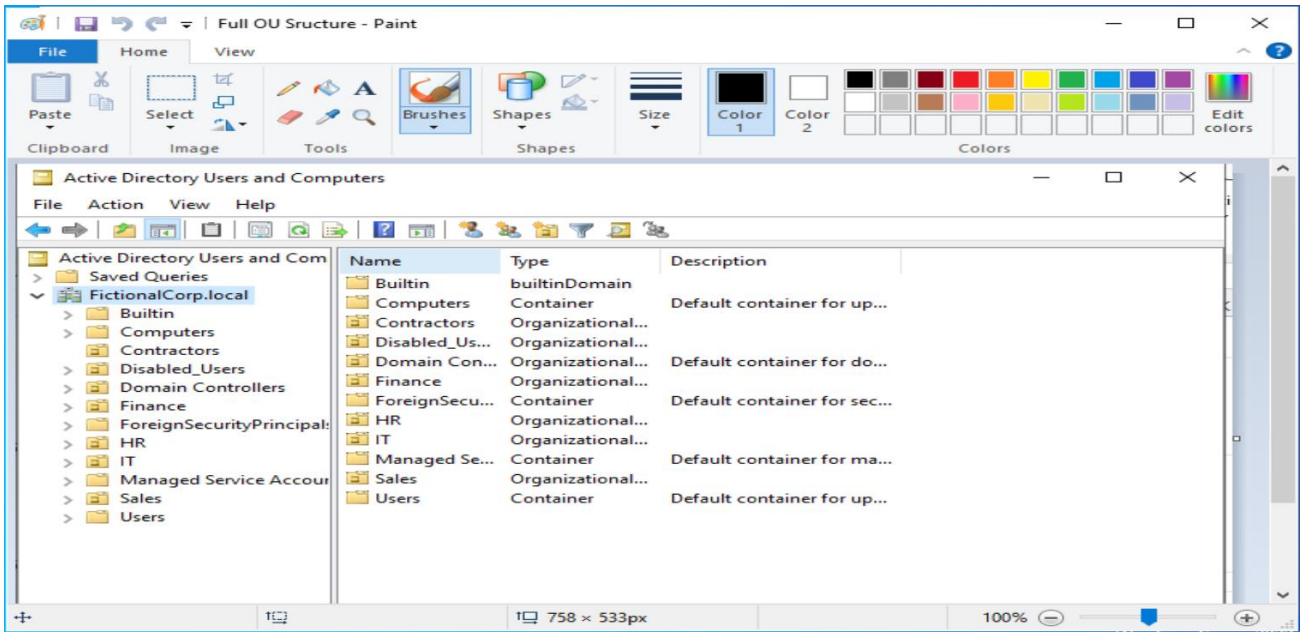
| Component | Value |
|---|---|
| Virtualization Platform | VMware Workstation |
| Operating System | Windows Server 2022 Standard Evaluation (Desktop Experience) |
| Server Name | DC01 |
| IP Address | 192.168.10.10 (Static) |
| Subnet Mask | 255.255.255.0 |
| Preferred DNS | 127.0.0.1 (Self-referencing — post-AD promotion) |
| Domain Name | FictionalCorp.local |
| NetBIOS Name | FICTIONALCORP |
| Network Adapter 1 | NAT — Internet access for Azure connectivity |
| Network Adapter 2 | Host-only — Private AD network (192.168.10.x) |
| DSRM Password | Stored securely in lab credentials vault |

### 2.2 Azure Connection Note

> Every user account and security group created in this lab will be synchronized to Microsoft Entra ID in Lab 05 using Azure AD Connect. The OU structure, naming conventions, and group design established here become the on-premises source of truth for the entire hybrid identity environment. Build it cleanly — a messy on-premises structure creates a messy cloud structure.

## 3. Organizational Unit Structure

The following OU structure was designed and implemented in Active Directory Users and Computers (ADUC) to reflect a realistic enterprise department hierarchy. OUs serve as containers that organize users by department, enabling targeted Group Policy application, delegated administration, and clean access reviews.

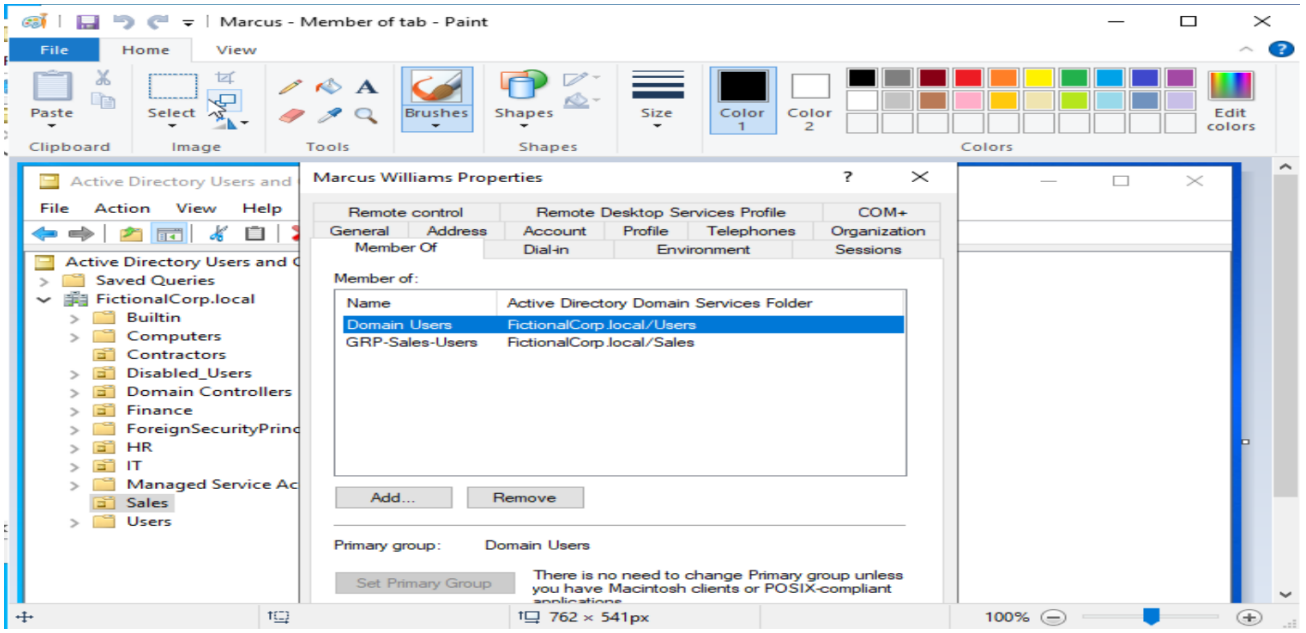



| Organizational Unit | Purpose | Contents |
|---|---|---|
| HR | Human Resources department users | Permanent employees — HR roles |
| Finance | Finance department users | Permanent employees — Finance roles + SoD groups |
| IT | Information Technology department users | Permanent employees + contractor accounts |
| Sales | Sales department users | Permanent employees — Sales roles |
| Disabled_Users | Terminated and inactive accounts | All disabled accounts — 30-day retention before deletion |
| Contractors | All contractor and temporary worker accounts | Time-limited accounts with automatic expiry dates |

> Design Decision: Separating contractors into a dedicated OU provides immediate visual identification during access reviews, simplifies Group Policy targeting, and isolates contractor accounts from permanent staff — a standard enterprise IAM practice.

## 4. Security Groups Created

Role-based security groups were created in each departmental OU. Access permissions are assigned to groups — never to individual users directly. This enforces the principle of least privilege and ensures access changes can be made at the role level without touching individual accounts.




| Group Name | OU Location | Scope | Type | Purpose |
|---|---|---|---|---|
| GRP-HR-Users | HR | Global | Security | Standard access for all HR staff |
| GRP-Finance-Users | Finance | Global | Security | Standard access for all Finance staff |
| GRP-Finance-InvoiceCreate | Finance | Global | Security | Permission to create invoices (SoD control) |
| GRP-Finance-InvoiceApprove | Finance | Global | Security | Permission to approve invoices (SoD control) |
| GRP-IT-Users | IT | Global | Security | Standard access for all IT staff |
| GRP-Sales-Users | Sales | Global | Security | Standard access for all Sales staff |

**SOD DESIGN NOTE**

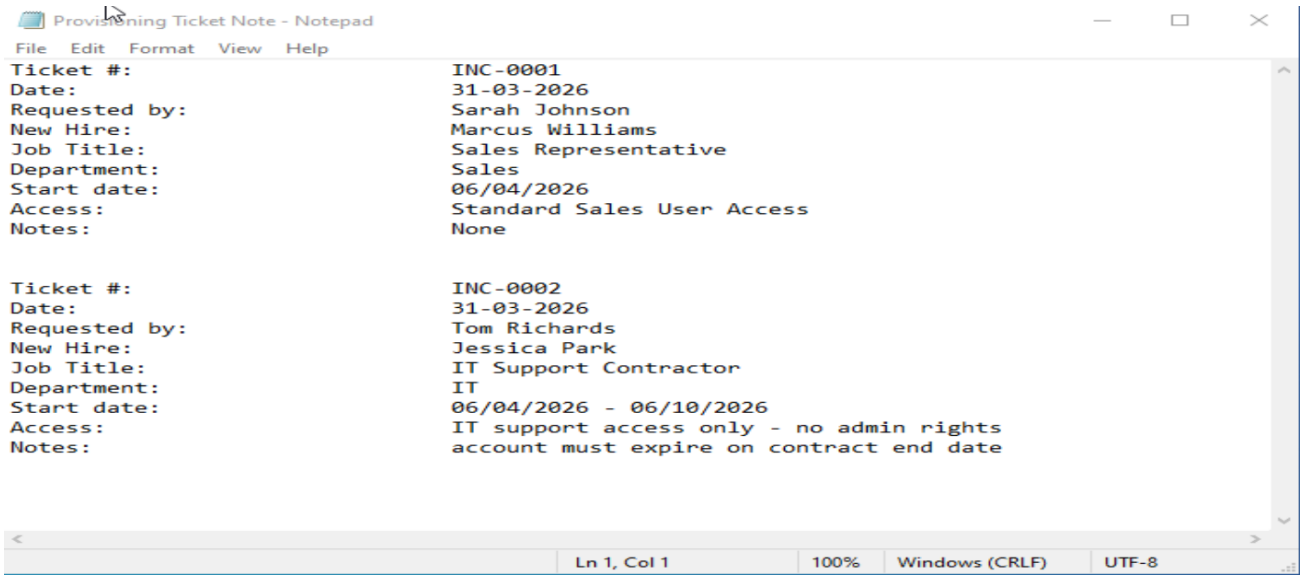
SoD Note: GRP-Finance-InvoiceCreate and GRP-Finance-InvoiceApprove were created specifically to enforce Segregation of Duties in Lab 14. No user should be a member of both groups simultaneously — this would allow a single person to create and approve their own invoices, a critical internal control violation.

## 5. Provisioning Tickets Processed

Two provisioning requests were processed during this lab, simulating the real ticket-based workflow used by enterprise IAM teams. Every action was documented in the provisioning log before being executed.

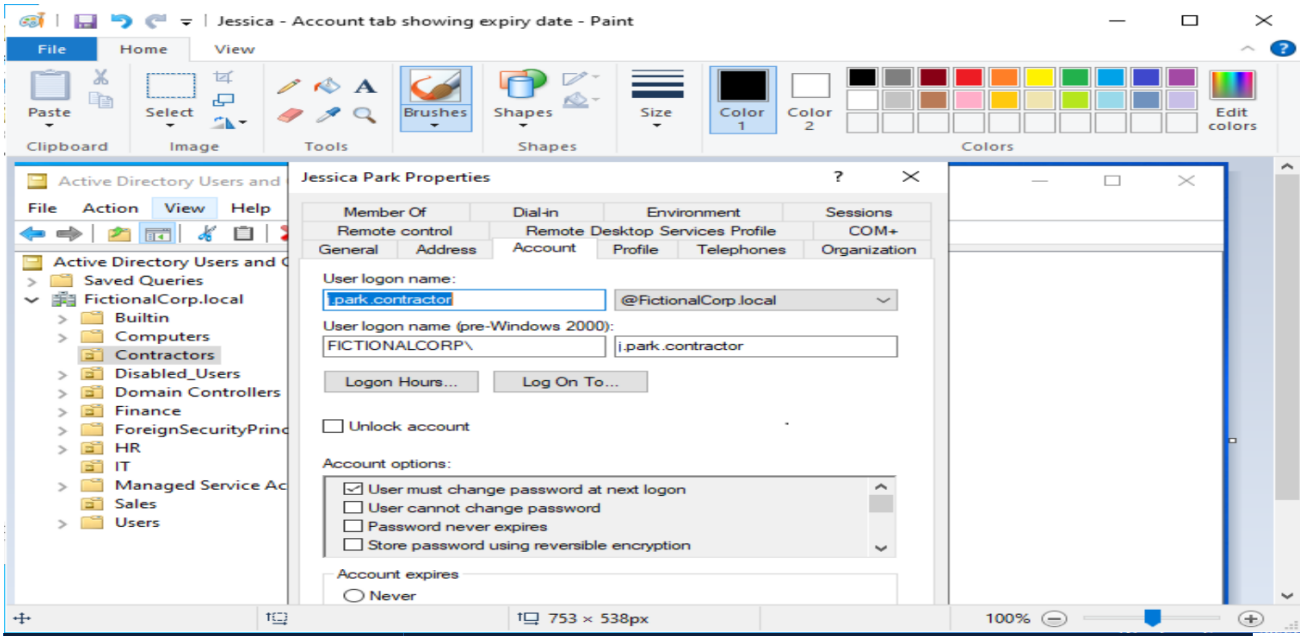
## 5.1 Ticket INC-0001 — New Permanent Employee

| Field | Value |
|---|---|
| Ticket Number | INC-0001 |
| Requested By | Sarah Johnson — HR Manager |
| Full Name | Marcus Williams |
| Job Title | Sales Representative |
| Department | Sales |
| Account Type | Permanent Employee |
| Username | m.williams |
| OU Placement | FictionalCorp.local/Sales |
| Groups Assigned | GRP-Sales-Users |
| | |
| Account Expiry | None — permanent employee |
| Password Policy | User must change password at next logon |




> Least Privilege Applied: Marcus was assigned only to GRP-Sales-Users. He was not added to Finance, HR, or IT groups. Access is granted at the minimum level required for his job function.

## 5.2 Ticket INC-0002 — New Contractor




| Field | Value |
|---|---|
| Ticket Number | INC-0002 |
| Requested By | Tom Richards — IT Manager |
| Full Name | Jessica Park |
| Job Title | IT Support Contractor |
| Department | IT |
| Account Type | Contractor — Time-limited |
| Username | j.park.contractor |
| OU Placement | FictionalCorp.local/Contractors |
| Groups Assigned | GRP-IT-Users |
| Account Expiry | 90 days from provisioning date (auto-disables) |
| Password Policy | User must change password at next logon |

Contractor Control: The .contractor suffix in the username instantly identifies this as a contractor account during any AD search or audit. The 90-day account expiry means access is automatically revoked when the contract ends — no manual intervention required, no risk of forgotten access lingering past the contract end date.

## 6. Key Design Decisions & Rationale

### Why Host-only + NAT dual adapter configuration?

A single Host-only adapter isolates the VM on a private network with no internet access. Adding a NAT adapter alongside it provides internet connectivity required for downloading Azure AD Connect in Lab 05 and connecting to the Azure tenant. This mirrors how production domain controllers are segmented — on a private management network with controlled external access.

### Why 127.0.0.1 as the Preferred DNS?

After AD DS installation and domain promotion, the server becomes its own DNS server. Setting 127.0.0.1 as the preferred DNS before promotion ensures the server resolves DNS queries against itself after the role is installed. Using an external DNS address would break AD functionality.

### Why separate Contractors OU from department OUs?

Placing contractors in a dedicated OU provides three operational advantages: (1) visual isolation during access reviews makes contractor accounts immediately identifiable, (2) Group Policies specific to contractors can be applied at the OU level, and (3) when Azure AD Connect syncs users to Entra ID in Lab 05, the OU path appears in the cloud and confirms the account type without additional attributes.

### Why never assign permissions to individual user accounts?

Assigning permissions to individual users creates an unmanageable access environment at scale. If a user changes roles, their permissions must be individually audited and updated. With role-based groups, a role change is a single group membership update that immediately adjusts all associated permissions. This is a foundational IAM principle that every interviewer will probe.

## 7. Interview Story

**INTERVIEW STORY — LAB 01**

When asked: “Walk me through your provisioning process.” Answer: “When I receive a provisioning request, I work from the ticket — I need the full name, job title, department, and whether they are a permanent employee or contractor. I create the account in the correct OU based on their department, assign them to the role-based security group for that department following least privilege, and set the password to force a change on first login. For contractors, I always set an account expiry matching the contract end date — access revokes automatically without relying on anyone to remember it. Every action is logged in our provisioning tracker with the ticket number for the audit trail. In a hybrid environment, these same users then sync to Entra ID via Azure AD Connect, so cloud access follows automatically.”